

Machine Learning-Based Analysis of Corporate Tax Avoidance: Evidence from Korean Listed Companies

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ABSTRACT

Abstract

Corporate tax avoidance has become a central concern for regulators, investors, and researchers due to its direct implications for fiscal sustainability, financial reporting quality, and corporate governance. Despite growing interest in applying machine learning to tax research, most existing studies address a single analytical task under methodological limitations such as random train-test splitting, which introduces temporal information leakage when panel data are used. This study proposes a comprehensive, multi-task analytical framework that addresses these limitations by integrating regression, classification, unsupervised clustering, and explainable artificial intelligence within a unified leakage-free experimental design. Using the Korean Tax Avoidance Panel (KoTaP) dataset—comprising 12,653 firm-year observations from 1,754 non-financial listed companies over the period 2011–2024—we employ a one-year-ahead forecasting protocol in which all predictive models are trained exclusively on historically available information. For regression, a LightGBM model with hyperparameter optimisation via Optuna achieves the best predictive performance ($R^2 = 0.108$, MAE = 0.149, RMSE = 0.209) relative to a linear baseline and a residual neural network. For classification, an optimised Random Forest classifier achieves 58.99% accuracy with a weighted ROC-AUC of 0.724 in identifying firms' future tax-risk categories. Unsupervised analysis using Gaussian Mixture Models reveals meaningful firm heterogeneity driven primarily by size, foreign ownership, and asset structure. SHAP analysis further indicates that historical tax behaviour is highly informative for future corporate tax avoidance predictions. The findings carry practical implications for tax authorities, auditors, investors, and financial analysts seeking to develop data-driven tax-risk screening approaches.

Keywords: corporate tax avoidance; machine learning; LightGBM; random forest; SHAP; leakage-free prediction; KoTaP; Korean listed firms; clustering.

Contents

1	Introduction	4
2	Literature Review	5
2.1	Corporate Tax Avoidance: Measurement and Determinants	5
2.2	Machine Learning Applications in Tax Avoidance Research	6
2.3	Leakage-Free Evaluation in Panel Data Settings	6
2.4	Explainable Artificial Intelligence in Financial Analytics	6
3	Data	7
3.1	Dataset Description	7
3.2	Variable Taxonomy	7
3.3	Data Quality Assessment	8
3.4	Descriptive Statistics	8
3.5	Temporal Trends in Tax Avoidance	8
3.6	Cross-Sectional Variation by Firm Characteristics	9
4	Methodology	11
4.1	Analytical Framework Overview	11
4.2	Feature Selection	11
4.3	Leakage-Free Prediction Framework	12
4.4	Classification Label Construction	12
4.5	Regression Models	13
4.5.1	Multiple Linear Regression	13
4.5.2	LightGBM	13
4.5.3	Residual MLP	13
4.6	Classification Models	13
4.6.1	Logistic Regression	13
4.6.2	Random Forest	14
4.6.3	Support Vector Machine	14
4.7	Unsupervised Clustering	14
4.7.1	K-Means Clustering	14

4.7.2	Gaussian Mixture Model	14
4.8	SHAP Explainability Analysis	14
4.9	Performance Metrics	15
5	Results	15
5.1	Regression Results	15
5.1.1	Multiple Linear Regression	15
5.1.2	LightGBM with Progressive Feature Engineering	16
5.1.3	Residual MLP	16
5.2	Classification Results	16
5.2.1	Logistic Regression	16
5.2.2	Optimised Random Forest	16
5.2.3	Support Vector Machine	17
5.3	Clustering Results	17
5.3.1	K-Means	17
5.3.2	Gaussian Mixture Model	17
5.4	SHAP Explainability Results	18
5.5	Overall Findings	19
6	Discussion	19
6.1	Predictability of Corporate Tax Avoidance	19
6.2	Tax-Risk Classification and Its Practical Implications	19
6.3	Firm Heterogeneity and Clustering	20
6.4	Explainability and Financial Interpretation	20
6.5	Methodological Implications	20
6.6	Limitations	21
7	Practical Applications and Implications	21
8	Conclusion	22

1 Introduction

1 Introduction Corporate tax avoidance refers to a broad set of legally permissible strategies employed by firms to reduce their effective tax burden below the statutory rate. Although these strategies do not violate tax law, aggressive tax planning raises substantial concerns regarding financial reporting transparency, corporate governance quality, and long-term fiscal sustainability [8]. Governments face material revenue losses as a result, while investors and regulators struggle to assess the associated financial risks from externally available disclosures alone.

The quantitative study of corporate tax avoidance has traditionally relied on regression-based econometric models that examine the relationship between firm characteristics and effective tax rates [6]. These approaches offer interpretability but are inherently limited in their ability to capture complex, non-linear interactions among the many financial, governance, and operational variables that jointly determine a firm’s tax position. The advent of large-scale panel datasets and modern machine learning algorithms has created new opportunities to address these limitations [7].

Despite this progress, several methodological gaps persist in the literature. First, the majority of existing studies treat tax avoidance prediction as a single analytical task—either regression or classification—without providing a comprehensive evaluation of multiple techniques within a unified experimental setting. Second, many prior studies rely on random train-test splitting strategies that may introduce temporal information leakage when panel data are analysed, yielding prediction performance estimates that are systematically over-optimistic and unlikely to generalise to future observations [14]. Third, the explainability of machine learning predictions has received limited attention in the tax avoidance literature, making it difficult for practitioners to understand which financial variables drive model outputs.

To address these limitations, this study develops a comprehensive multi-task analytical framework applied to the Korean Tax Avoidance Panel (KoTaP) dataset [11]. Korea constitutes a particularly instructive setting for this investigation: the mandatory adoption of Korean International Financial Reporting Standards (K-IFRS) in 2011 substantially improved the consistency and comparability of financial reporting across listed firms and over time, enabling the construction of a long-horizon research-grade panel that supports both retrospective analysis and forward-looking prediction [13].

The proposed framework encompasses four complementary analytical components. First, regression models are developed to predict one-year-ahead corporate tax avoidance using a leakage-free chronological data-splitting protocol that eliminates temporal information contamination. Second, classification models assign firms to discrete tax-risk categories, providing directly interpretable outputs for risk screening. Third, unsupervised clustering techniques are applied to identify latent firm segments with similar financial and governance profiles. Fourth, SHAP (SHapley Additive exPlanations) analysis is employed to interpret the variable-level contributions to the predictions of the selected machine learning model, bridging the gap between predictive performance and financial interpretation.

The contributions of this study are as follows:

1. A leakage-free multi-task machine learning framework for corporate tax avoidance analysis, integrating regression, classification, clustering, and explainability within a single consistent experimental design.
2. Empirical evidence that historical tax behaviour provides substantial predictive information about future corporate tax avoidance.
3. A systematic comparison of machine learning and neural network models under identical evaluation conditions, including hyperparameter optimisation via Optuna and Randomized Search Cross-Validation.
4. Firm-level cluster profiles that reveal the financial and governance dimensions associated with heterogeneous tax behaviour among Korean listed companies.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the dataset and descriptive characteristics. Section 4 presents the analytical framework and modelling approach. Section 5 reports the empirical results. Section 6 discusses the findings and their implications. Section 7 presents the practical applications and implications. Section 8 concludes the study.

2 Literature Review

2.1 Corporate Tax Avoidance: Measurement and Determinants

Corporate tax avoidance is inherently latent—it cannot be observed directly from publicly available disclosures because taxable income and detailed tax adjustments are not fully disclosed [8]. Empirical research therefore relies on standardised proxy measures derived from financial statements. The two most widely used proxies are the GAAP Effective Tax Rate (GETR), defined as total income tax expense divided by pre-tax income, and the Cash Effective Tax Rate (CETR), defined as cash taxes paid divided by pre-tax income [5]. GETR captures the accrual-based tax burden and reflects deferred tax effects, whereas CETR isolates actual cash outflows and is less susceptible to deferred tax timing differences. A lower effective tax rate, relative to the statutory rate, is conventionally interpreted as evidence of more aggressive tax planning [6].

The determinants of corporate tax avoidance have been extensively studied. Firm size is among the most debated variables: the political cost hypothesis suggests that larger firms face greater regulatory scrutiny and therefore exhibit higher effective tax rates, whereas the resource hypothesis argues that larger firms possess greater capacity to engage in sophisticated tax planning [8]. Leverage has been found to affect the tax burden through the tax deductibility of interest payments [4]. Profitability is positively associated with tax expense in absolute terms but may be negatively associated with effective tax rates when profitable firms have greater

capacity to exploit tax incentives [5]. Corporate governance mechanisms, including the quality of external auditors and the composition of ownership, have also been shown to moderate tax avoidance behaviour [6].

2.2 Machine Learning Applications in Tax Avoidance Research

The application of machine learning to corporate tax avoidance has grown substantially in the past decade. Guenther et al. [7] demonstrate that machine learning models can forecast one-year-ahead effective tax rates from accounting disclosures with greater accuracy than simple mean-reversion benchmarks, although predictive performance remains moderate. Borrotti et al. [2] employ machine learning to identify European corporate groups at risk of aggressive tax-location strategies using publicly available accounting information, highlighting the potential of prediction-based approaches for proactive regulatory enforcement.

Rahman et al. [12] evaluate logistic regression, decision trees, and random forests for tax avoidance classification using Malaysian listed-firm data, finding that the choice of feature sets—including industry, governance, and firm characteristics—materially affects predictive reliability. Collectively, these studies establish the feasibility of machine learning-based tax avoidance screening but also highlight persistent challenges related to feature design, evaluation protocol design, and the interpretability of model outputs.

2.3 Leakage-Free Evaluation in Panel Data Settings

A critical but frequently overlooked methodological issue in machine learning applications to panel data concerns temporal information leakage. Random train-test splitting permits future information to inform model training when observations from the same firm appear in both splits. This produces performance estimates that may be overly optimistic and may not generalise to genuinely out-of-sample future periods [14].

Song and Kim [14] formalise a leakage-free evaluation protocol for one-year-ahead forecasting of Korean tax-avoidance proxies using the KoTaP dataset, enforcing chronological data splits and a fit-on-train-only preprocessing principle. Their study focuses on an ablation comparison between raw and derived feature configurations using gradient boosting and TabTransformer models. The present study adopts the same leakage-free principle while broadening the analytical scope to encompass classification, clustering, and SHAP-based explainability.

2.4 Explainable Artificial Intelligence in Financial Analytics

The opacity of ensemble and neural network models has motivated growing interest in explainable artificial intelligence (XAI) methods that attribute predictions to individual input features. SHAP [10] provides a theoretically grounded approach based on cooperative game theory, assigning each feature a contribution value with desirable properties including local accuracy and consistency. In financial analytics, SHAP has been applied to credit risk, bankruptcy prediction, and fraud

detection, while its application to corporate tax avoidance remains comparatively limited. This study employs SHAP to interpret the final optimised LightGBM regression model, thereby connecting predictive performance to economically meaningful financial variables.

3 Data

3.1 Dataset Description

The empirical analysis is based on the Korean Tax Avoidance Panel (KoTaP) dataset [11], a publicly available research-grade panel that standardises firm-year financial, governance, and tax-related information for non-financial companies listed on the Korea Exchange. The dataset covers both the KOSPI and KOSDAQ market segments over the period 2011–2024, spanning the post-K-IFRS adoption era and thereby supporting consistent accounting definitions across the observation window.

The final sample comprises 12,653 firm-year observations from 1,754 unique firms across 14 fiscal years, encoded in 65 variables. The panel is unbalanced because firms enter and exit the sample due to incorporation, listing, and delisting events. Source data are reconstructed from machine-readable disclosures filed through Korea’s electronic disclosure system (DART), which operates an XBRL-based reporting framework aligned with K-IFRS.

3.2 Variable Taxonomy

Following the KoTaP documentation, variables are organised into the groups summarised in Table 1.

Table 1: Variable taxonomy of the KoTaP dataset.

Group	Variables
Identifiers / Meta	name, stock, year, KOSPI, fnd_year, fiscal, ind
Raw (directly extracted)	big4, forn, own, c_asset, inv, asset, sales, cogs, dep, tax, rec, ni, ocf, cash, tan, land, cip, intan, liab, c_liab, pti, total, equit
Derived (ratios / indicators)	SIZE, LEV, CUR, GRW, ROA, ROE, CFO, PPE, AGE, INVREC, MB, TQ, LOSS
Derived (lag features)	lag_asset, lag_liab, lag_equit, lag_sales, lag_total, lag1_ni, lag_c_asset, lag_c_liab
Tax-avoidance proxies (targets)	CETR, GETR, CETR3, GETR3, CETR5, GETR5, TSTA, TSDA
Tax-avoidance proxies (auxiliary)	A_CETR, A_GETR, A_CETR3, A_GETR3, A_CETR5, A_GETR5

3.3 Data Quality Assessment

A primary data quality assessment was conducted prior to any modelling activity. The dataset contains no missing values across all 65 variables and no duplicate records, with each observation representing a unique firm-year instance. This completeness eliminates the need for imputation and supports the integrity of the temporal panel structure.

Outlier analysis using the interquartile range (IQR) method identified elevated proportions of extreme observations in several raw financial variables, most notably construction in progress (cip: 15.41%), depreciation (dep: 14.68%), total deferred tax (lag_total: 14.17%), and operating cash flows (ocf: 13.76%). These observations reflect genuine economic heterogeneity—particularly variation in firm size and capital intensity—rather than measurement errors. They were therefore retained throughout the analysis to preserve the economic characteristics of the sample.

3.4 Descriptive Statistics

Table 2 reports descriptive statistics for the key analytical variables. The sample exhibits substantial cross-sectional variation in firm size, profitability, leverage, and tax burden, consistent with the broad representation of KOSPI and KOSDAQ firms across diverse industries.

Table 2: Descriptive statistics of key variables (N = 12,653).

Variable	Mean	Std. Dev.	Min	Median	Max
CETR	0.2445	0.2197	0.0000	0.2100	1.0000
GETR	0.2040	0.1318	0.0000	0.1960	1.0000
SIZE	26.07	1.548	21.35	25.92	32.15
LEV	0.326	0.178	0.000	0.315	0.932
ROA	0.034	0.071	-0.651	0.028	0.512
ROE	0.053	0.162	-2.183	0.054	1.847
CFO	0.058	0.073	-0.485	0.052	0.531
GRW	0.079	0.261	-0.891	0.045	3.247
CUR	2.891	3.142	0.101	1.873	52.16
MB	1.124	1.283	0.052	0.742	18.95
TQ	1.087	0.814	0.251	0.843	11.72
AGE	27.13	15.84	1.000	24.00	124.0

3.5 Temporal Trends in Tax Avoidance

Table 3 documents the annual evolution of tax-avoidance proxies over the sample period. CETR exhibits an increasing trend from 0.2375 in 2011 to a peak of 0.2839 in 2019, followed by a notable decline to 0.1829 in 2021. GETR displays a comparatively more stable trajectory, fluctuating within a narrower band across the full sample period.

Table 3: Annual averages of tax-avoidance proxy measures, 2011–2024.

Year	CETR	GETR	CETR3	GETR3	CETR5	GETR5
2011	0.2375	0.2060	0.0885	0.1837	0.0938	0.1995
2012	0.2304	0.1916	0.1967	0.1916	0.1313	0.1910
2013	0.2447	0.2072	0.2122	0.2000	0.1586	0.1893
2014	0.2463	0.2085	0.2233	0.1997	0.1992	0.1965
2015	0.2525	0.2109	0.2372	0.2088	0.2217	0.2057
2016	0.2605	0.2130	0.2320	0.2132	0.2222	0.2031
2017	0.2705	0.2090	0.2344	0.2068	0.2334	0.2101
2018	0.2675	0.2185	0.2433	0.2022	0.2267	0.2068
2019	0.2839	0.2226	0.2420	0.2082	0.2291	0.2037
2020	0.2437	0.2131	0.2545	0.2234	0.2280	0.2104
2021	0.1829	0.1982	0.2495	0.2172	0.2143	0.2111
2022	0.2768	0.1972	0.1872	0.2026	0.2141	0.2032
2023	0.2433	0.1701	0.2047	0.1885	0.2114	0.1917
2024	0.1906	0.1751	0.2328	0.1827	0.1926	0.1799

3.6 Cross-Sectional Variation by Firm Characteristics

To motivate the subsequent predictive models, Tables 4–9 examine how tax-avoidance proxies vary across key firm-level dimensions.

Audit quality. Table 4 reveals that firms audited by Big Four accounting firms exhibit systematically higher effective tax rates ($\text{CETR} = 0.2571$) than non-Big Four firms ($\text{CETR} = 0.2315$). This pattern is consistent with the view that high-quality auditors constrain aggressive tax reporting, reducing the gap between book and taxable income.

Table 4: Tax-avoidance proxies by audit quality (Big4 indicator).

Big4	CETR	CETR3	CETR5	GETR	GETR3	GETR5
0 (Non-Big4)	0.2315	0.2104	0.1936	0.1903	0.1919	0.1912
1 (Big4)	0.2571	0.2260	0.2069	0.2138	0.2123	0.2092

Firm size. Table 5 shows a monotonic increase in CETR across size quartiles, rising from 0.1921 in the smallest quartile to 0.2755 in the largest. This evidence is consistent with the political cost hypothesis, under which larger firms face greater institutional scrutiny.

Table 5: Tax-avoidance proxies by firm size quartile (SIZE).

SIZE Quartile	CETR	CETR3	CETR5	GETR	GETR3	GETR5
Q1 (Smallest)	0.1921	0.1701	0.1562	0.1605	0.1589	0.1604
Q2	0.2382	0.2145	0.1954	0.1986	0.1981	0.1975
Q3	0.2703	0.2424	0.2198	0.2183	0.2185	0.2135
Q4 (Largest)	0.2755	0.2451	0.2291	0.2299	0.2320	0.2286

Profitability. The most economically striking pattern emerges from Table 6: firms in the lowest ROA quartile exhibit a CETR of 0.3981, nearly 2.5 times that of the highest ROA

quartile (0.1555). This counter-intuitive result may reflect that low-profitability firms can have temporarily inflated tax ratios because of small pre-tax-income denominators, while highly profitable firms may have greater capacity to deploy tax minimisation strategies.

Table 6: Tax-avoidance proxies by profitability quartile (ROA).

ROA Quartile	CETR	CETR3	CETR5	GETR	GETR3	GETR5
Q1 (Lowest ROA)	0.3981	0.2773	0.2569	0.2640	0.2211	0.2224
Q2	0.2296	0.2230	0.2057	0.1821	0.2021	0.1977
Q3	0.1930	0.1978	0.1821	0.1820	0.1971	0.1932
Q4 (Highest ROA)	0.1555	0.1740	0.1557	0.1790	0.1872	0.1865

Leverage. Table 7 indicates a mild positive association between leverage and effective tax rates, with CETR rising from 0.2159 in the lowest leverage quartile to 0.2561 in the highest. This finding may reflect differences in firm maturity and stability or the influence of other balance-sheet characteristics.

Table 7: Tax-avoidance proxies by leverage quartile (LEV).

LEV Quartile	CETR	CETR3	CETR5	GETR	GETR3	GETR5
Q1 (Lowest LEV)	0.2159	0.2118	0.1836	0.1789	0.1782	0.1756
Q2	0.2468	0.2146	0.1960	0.1964	0.2021	0.1980
Q3	0.2574	0.2162	0.2004	0.2087	0.2047	0.2014
Q4 (Highest LEV)	0.2561	0.2296	0.2205	0.2233	0.2226	0.2248

Foreign ownership. Table 8 documents a positive association between foreign ownership concentration and effective tax rates. Firms in the highest foreign ownership quartile (CETR = 0.2619) exhibit a higher average CETR than firms in the lowest quartile (0.2304), consistent with the possibility that stronger external monitoring is associated with less aggressive tax planning.

Table 8: Tax-avoidance proxies by foreign ownership quartile (FORN).

FORN Quartile	CETR	CETR3	CETR5	GETR	GETR3	GETR5
Q1 (Lowest)	0.2304	0.2002	0.1818	0.1854	0.1853	0.1856
Q2	0.2414	0.2231	0.2062	0.1985	0.2010	0.1980
Q3	0.2425	0.2185	0.2037	0.2001	0.2043	0.2012
Q4 (Highest)	0.2619	0.2305	0.2089	0.2233	0.2171	0.2152

Ownership concentration. Table 9 shows a non-monotonic relationship between ownership concentration and effective tax rates. CETR rises from Q1 (0.2360) to Q3 (0.2498) before declining slightly in Q4 (0.2479), suggesting that the association between ownership concentration and tax behaviour may be non-linear.

Table 9: Tax-avoidance proxies by ownership concentration quartile (OWN).

OWN Quartile	CETR	CETR3	CETR5	GETR	GETR3	GETR5
Q1 (Lowest)	0.2360	0.1955	0.1819	0.1945	0.1875	0.1877
Q2	0.2425	0.2189	0.2029	0.1994	0.2021	0.2015
Q3	0.2498	0.2321	0.2093	0.2094	0.2134	0.2072
Q4 (Highest)	0.2479	0.2256	0.2063	0.2039	0.2046	0.2036

4 Methodology

4.1 Analytical Framework Overview

This study adopts a quantitative, data-driven analytical framework that integrates five complementary components: (1) exploratory data analysis and feature selection, (2) leakage-free regression modelling for one-year-ahead prediction, (3) supervised classification of tax-risk categories, (4) unsupervised clustering for firm segmentation, and (5) SHAP-based explainability analysis. All predictive components share a common leakage-free experimental protocol that preserves the temporal ordering of observations and prevents future information from entering the training pipeline.

4.2 Feature Selection

Starting from the 65 variables in KoTaP, the input feature set was constructed through a three-stage selection procedure. First, correlation analysis among candidate predictors was examined to identify redundant variables, revealing strong collinearity between ROA and ROE, and between MB and TQ. Second, Variance Inflation Factors (VIF) were computed to formally assess multicollinearity. Variables directly involved in the construction of tax-avoidance proxy measures—including pre-tax income (pti), net income (ni), total tax expense (tax), operating cash flows (ocf), total assets, liabilities, and equity—were excluded from the principal predictive feature set to reduce direct target construction overlap. Third, a Random Forest importance ranking was applied to the remaining candidates, with profitability (ROA), growth (GRW), and operating cash flow (CFO) receiving comparatively high predictive importance.

The final feature set for the regression and classification models comprises 17 explanatory variables: big4, forn, own, SIZE, LEV, CUR, GRW, ROA, CFO, cash, PPE, tan, intan, AGE, INVREC, TQ, and LOSS. In extended experiments, auxiliary tax-history variables—including CETR3, GETR3, CETR5, GETR5, A_CETR, A_GETR, A_CETR3, A_GETR3, A_CETR5, and A_GETR5—are added as additional inputs, all restricted to values observable at the prediction year t .

4.3 Leakage-Free Prediction Framework

The prediction task is formulated as a one-year-ahead forecasting problem: for each firm i observed at year t , the objective is to predict the tax-avoidance proxy $Y_{i,t+1}$ using only covariates $X_{i,t}$ observable at year t . This ex-ante setting mirrors the operational constraints faced by regulators and investors who must assess firms before future tax disclosures become available.

To enforce leakage prevention, the dataset is divided chronologically as shown in Table 10.

Table 10: Leakage-free chronological data split.

Split	Input years (t)	Target year (t+1)	Observations
Training	2011–2022	2012–2023	9,970
Test	2023	2024	929
Total	2011–2023	2012–2024	10,899

All preprocessing transformations—including feature scaling, outlier treatment, and derived feature construction—are fitted exclusively on the training split and applied without modification to the test split. Hyperparameter optimisation for selected models uses a six-fold expanding-window cross-validation strategy within the training period, as shown in Table 11.

Table 11: Six-fold expanding-window cross-validation folds.

Fold	Training years	Validation year
1	2011–2016	2017
2	2011–2017	2018
3	2011–2018	2019
4	2011–2019	2020
5	2011–2020	2021
6	2011–2021	2022

The one-year-ahead target $Y_{i,t+1}$ is constructed by shifting each firm’s CETR forward by one period within the panel, using the firm identifier to ensure that the shift respects firm boundaries. Firm-years for which no subsequent observation exists are excluded, yielding 1,754 dropped observations and a final predictive dataset of 10,899 firm-year pairs.

4.4 Classification Label Construction

For the classification task, firms are assigned to one of three tax-risk categories—High, Medium, and Low—based on year-specific quartiles of the one-year-ahead CETR:

$$TaxRisk_{i,t+1} = \begin{cases} \text{High,} & CETR_{i,t+1} \leq Q1_t, \\ \text{Medium,} & Q1_t < CETR_{i,t+1} < Q3_t, \\ \text{Low,} & CETR_{i,t+1} \geq Q3_t, \end{cases} \quad (1)$$

where $Q1_t$ and $Q3_t$ denote the first and third quartiles of $CETR_{t+1}$ computed separately for each fiscal year. This dynamic labelling strategy accounts for annual variation in the aggregate distribution of tax avoidance and maintains approximately balanced class proportions: High ($n = 2,730$; 25%), Medium ($n = 5,439$; 50%), and Low ($n = 2,730$; 25%).

4.5 Regression Models

Three regression models are evaluated to predict the continuous one-year-ahead CETR.

4.5.1 Multiple Linear Regression

Multiple Linear Regression (MLR) serves as the statistical baseline, establishing the performance of a purely linear specification under the same leakage-free protocol.

4.5.2 LightGBM

LightGBM [9] is a gradient boosted decision tree implementation that combines histogram-based split finding with leaf-wise tree growth. It is selected as the primary machine learning model because of its ability to capture non-linear relationships and complex feature interactions in financial tabular data. Hyperparameter optimisation is performed using the Optuna framework [1] with 50 trials minimising the average validation RMSE across the six expanding-window folds. The search space encompasses the number of estimators (200–800), learning rate (0.01–0.10), maximum depth (3–10), number of leaves (15–127), minimum child samples (10–50), subsample ratio (0.7–1.0), and column sampling ratio (0.7–1.0).

4.5.3 Residual MLP

A Residual Multi-Layer Perceptron (MLP) is evaluated as a deep learning comparator. The architecture consists of two hidden layers (128 and 64 units) with ReLU activation, trained using the Adam optimiser with a learning rate of 0.001, batch size of 64, and early stopping on validation loss. Input features are standardised using a StandardScaler fitted on the training split only.

4.6 Classification Models

Three supervised classifiers are trained on the same feature set and data split to predict future tax-risk categories.

4.6.1 Logistic Regression

Logistic Regression serves as the statistical classification baseline, trained with standardised inputs and a multinomial objective function with a maximum of 1,000 iterations.

4.6.2 Random Forest

A Random Forest classifier [3] is selected as the primary ensemble model. Baseline performance is first established using default hyperparameters, followed by optimisation via Randomized Search Cross-Validation. The search explores the number of estimators (100–300), maximum depth ($\{\text{None}, 10, 20\}$), minimum samples per split (2–7), minimum samples per leaf (1–4), maximum features ($\{\text{sqrt}, \log 2\}$), and class weighting ($\{\text{None}, \text{balanced}\}$).

4.6.3 Support Vector Machine

A Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel is evaluated as a third comparator. Both a baseline configuration ($C = 1.0$, $\gamma = \text{scale}$) and an optimised configuration obtained via Randomized Search are assessed.

4.7 Unsupervised Clustering

Clustering analysis is applied to the full 17-variable feature set to identify naturally occurring firm segments. All features are standardised using RobustScaler prior to clustering to reduce the influence of skewed distributions and extreme values.

4.7.1 K-Means Clustering

K-Means clustering is applied to a cleaned dataset from which 127 anomalous observations (1.0% of the sample) identified by the Isolation Forest algorithm are removed. The number of clusters is evaluated for $K \in \{2, \dots, 10\}$ using the Elbow Method and Silhouette Scores.

4.7.2 Gaussian Mixture Model

Given the sensitivity of K-Means to imbalance and overlapping distributions, a Gaussian Mixture Model (GMM) is evaluated as an alternative. GMM models clusters as probability distributions and allows observations to exhibit overlapping cluster memberships. The number of components is evaluated using the Bayesian Information Criterion (BIC).

4.8 SHAP Explainability Analysis

SHAP (SHapley Additive exPlanations) [10] is applied only to the optimised LightGBM regression model to quantify variable-level contributions to one-year-ahead CETR predictions. A TreeExplainer is used to obtain SHAP values. Global feature importance is summarised as the mean absolute SHAP value across the analysed observations, while dependence analysis is used to inspect the relationship between the most influential feature and the model output. SHAP values are treated as model-based contributions rather than causal effects.

4.9 Performance Metrics

Regression performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2):

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|, \quad (2)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}, \quad (3)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (4)$$

Classification performance is evaluated using Accuracy, weighted Precision, weighted Recall, weighted F1-score, and the weighted multiclass ROC-AUC score. Clustering quality is assessed using the Silhouette Score, while GMM model selection is additionally examined using BIC.

5 Results

5.1 Regression Results

5.1.1 Multiple Linear Regression

Table 12 presents the out-of-sample performance of the Multiple Linear Regression baseline on the 2023 test set when predicting 2024 CETR. The model yields an MAE of 0.1753, RMSE of 0.2320, and a negative R^2 of -0.0983, indicating that a constant-mean predictor would outperform the linear model on this test set. Initial experiments demonstrated that the model performed better with the expanded financial feature set than with the extended tax-history variables, because the addition of highly correlated tax-history proxies increased redundancy without improving predictive accuracy under a linear specification.

Table 12: Regression model comparison—test set performance (predicting $CETR_{t+1}$).

Model	Feature Configuration	MAE	RMSE	R^2
Multiple Linear Regression	Expanded financial variables	0.1753	0.2320	-0.0983
LightGBM	Initial set (17 features)	0.1724	0.2229	-0.0145
LightGBM	Expanded set (36 features)	0.1687	0.2194	0.0176
LightGBM	Full financial set	0.1651	0.2178	0.0315
LightGBM (Optuna)	Full features + tax-history	0.1490	0.2090	0.1080
Residual MLP	Full features + tax-history	0.1606	0.2291	-0.0718

5.1.2 LightGBM with Progressive Feature Engineering

The LightGBM model is evaluated across four progressive feature configurations to isolate the incremental contribution of each variable group. As shown in Table 12, predictive accuracy improves as the feature space is expanded. The largest single improvement is achieved by incorporating auxiliary tax-history variables: MAE decreases by approximately 0.016 relative to the full financial feature set, while R^2 increases from 0.0315 to approximately 0.108. This pattern indicates that historical tax behaviour provides information about future avoidance that is not captured by contemporaneous financial ratios alone.

5.1.3 Residual MLP

The Residual MLP achieves an MAE of 0.1606 and RMSE of 0.2291—marginally better in MAE than the linear baseline but inferior to the optimised LightGBM model. The negative R^2 (-0.0718) indicates that the neural network struggles to generalise under the available sample size. Residual analysis also indicates that both the MLR and Residual MLP tend to concentrate predictions near the central range of the target distribution, underestimating extreme CETR values.

5.2 Classification Results

Table 13 summarises the performance of all classification models on the 929-observation test set.

Table 13: Classification model comparison—test set performance (tax-risk categories).

Model	Version	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	Baseline	0.5500	0.5300	0.5500	0.4800	0.6745
Random Forest	Baseline	0.5800	0.5600	0.5800	0.5300	0.7068
Random Forest	Optimised	0.5899	0.5722	0.5899	0.5640	0.7242
SVM (RBF)	Baseline	0.5800	0.6600	0.5800	0.5000	0.7203
SVM (RBF)	Optimised	0.5608	0.5276	0.5608	0.5092	0.6666

5.2.1 Logistic Regression

The Logistic Regression baseline achieves an accuracy of 55.0% and a weighted ROC-AUC of 0.675. Per-class analysis reveals strong performance for the Medium-risk category (Recall = 0.86) but near-zero recall for the Low-risk category (Recall = 0.06), indicating that linear decision boundaries are insufficient to separate low-avoidance firms from the dominant Medium class.

5.2.2 Optimised Random Forest

Hyperparameter optimisation yields consistent improvements across all major metrics. The optimised Random Forest achieves the highest overall accuracy (58.99%), F1-score (0.564), and

weighted ROC-AUC (0.724) among all evaluated classifiers. Per-class ROC-AUC analysis reveals that the model distinguishes High-risk firms most effectively (AUC = 0.849), while Low-risk and Medium-risk categories achieve moderate discrimination (AUC = 0.681 and 0.683, respectively). The improvement in F1-score relative to the baseline (+0.034) reflects a more balanced allocation of correct predictions across the three tax-risk categories.

The best hyperparameter configuration identified through Randomized Search is: $n_estimators = 230$, $max_depth = 20$, $max_features = \log 2$, $min_samples_leaf = 4$, and $class_weight = \text{balanced}$.

5.2.3 Support Vector Machine

The baseline SVM achieves the highest weighted precision (0.660) among all classifiers, indicating relatively reliable predictions when the model commits to a class. However, hyperparameter optimisation deteriorates SVM performance across most metrics, with accuracy declining by 1.92 percentage points and weighted ROC-AUC falling from 0.720 to 0.667. This suggests that the default RBF kernel parameterisation is better suited to the dataset characteristics than the configurations explored in the search space.

5.3 Clustering Results

5.3.1 K-Means

Prior to applying K-Means, the Isolation Forest algorithm identified 127 anomalous observations (1.0% of the sample), which were removed to prevent extreme values from distorting distance-based clustering. The Elbow Method suggested diminishing returns beyond $K = 5$, while Silhouette analysis yielded the highest score at $K = 2$ (Silhouette = 0.929), with decreasing scores for larger values of K .

Despite these diagnostics, the $K = 4$ solution was selected to examine potential firm sub-groups. The resulting cluster distribution is highly imbalanced: Cluster 0 contains 12,020 observations, while Clusters 1, 2, and 3 contain 86, 27, and 393 observations, respectively. PCA visualisation of the clustering solution indicates that the vast majority of firms are concentrated within a single dense region of the feature space.

5.3.2 Gaussian Mixture Model

To provide a complementary probabilistic segmentation, a Gaussian Mixture Model (GMM) was applied. The four-component solution was retained for interpretation based on the information-criterion analysis used in the study. The resulting cluster allocation is considerably more balanced than the K-Means solution.

Table 14: Clustering comparison: K-Means vs. Gaussian Mixture Model ($K = 4$).

Cluster	K-Means		GMM	
	N	%	N	%
0	12,020	96.0	5,766	46.0
1	86	0.7	5,507	44.0
2	27	0.2	96	0.8
3	393	3.1	1,157	9.2

The GMM produces a substantially more balanced allocation, with Clusters 0 and 1 each containing approximately 45% of observations and Cluster 3 capturing a meaningful sub-group of 9.2%. Cluster profiling identifies firm size (SIZE), foreign ownership (forn), cash holdings (cash), and tangibility (tan) as important dimensions differentiating the groups.

Table 15: Comparative assessment of clustering methods.

Criterion	K-Means	GMM
Cluster balance	Weak	Strong
PCA separation	Limited	Improved
Overlap handling	No	Yes
Interpretability	Moderate	High
Representation of firm heterogeneity	Weak	Strong

5.4 SHAP Explainability Results

SHAP analysis of the optimised LightGBM regression model identifies historical tax-avoidance variables as the dominant predictors of future CETR. The leading features by mean absolute SHAP value include A_GETR (0.0311), GETR5 (0.0242), pti (0.0103), CETR5 (0.0098), and TSDA (0.0082), followed by ROE, GRW, lag1_ni, A_CETR, and PPE.

Table 16: Top ten variables by mean absolute SHAP value for the optimised LightGBM regression model.

Feature	Mean absolute SHAP
A_GETR	0.031094
GETR5	0.024150
pti	0.010275
CETR5	0.009807
TSDA	0.008222
ROE	0.006354
GRW	0.004650
lag1_ni	0.004524
A_CETR	0.004159
PPE	0.003830

The SHAP dependence analysis shows a nonlinear relationship between `A_GETR` and the model output, supporting the earlier finding that nonlinear machine learning models can capture relationships that are difficult to represent through a simple linear specification. The SHAP findings complement the predictive results by demonstrating the strong contribution of historical tax-related information to future CETR predictions.

5.5 Overall Findings

Across the supervised learning experiments, the optimised LightGBM model produced the strongest regression performance, while the optimised Random Forest provided the strongest overall classification balance. The clustering analysis revealed meaningful firm heterogeneity despite the strong concentration of observations within a dominant financial profile, and the SHAP results provided an interpretable explanation of the variables contributing to future CETR predictions.

6 Discussion

6.1 Predictability of Corporate Tax Avoidance

The regression results reveal that one-year-ahead corporate tax avoidance is difficult to predict from financial statement data alone, even under a carefully designed leakage-free protocol. The optimised LightGBM model achieves an R^2 of approximately 0.105–0.108 depending on the exact reproducibility run, which is modest in absolute terms and consistent with the high variability of annual effective tax rates. The negative R^2 values obtained by the linear baseline and the neural network show that this predictive difficulty cannot be solved simply through a linear specification or by introducing additional neural-network complexity.

The most economically significant finding from the regression analysis is that historical tax-behaviour variables account for a substantial share of the predictive signal. The large performance improvement obtained after introducing tax-history variables indicates that firms’ past tax outcomes contain useful information about subsequent corporate tax behaviour. This is consistent with the temporal persistence hypothesis: firms’ tax strategies, legal structures, and governance arrangements may change more slowly than annual accounting outcomes.

6.2 Tax-Risk Classification and Its Practical Implications

The classification results suggest that machine learning models can provide useful but moderate discrimination among high-, medium-, and low-risk firms. The optimised Random Forest achieves a per-class AUC of 0.849 for the High-risk category, indicating that the model is more effective at identifying firms with future CETR in the lowest quartile than it is at distinguishing the less extreme categories.

The Low-risk category is consistently the hardest to classify, while the Medium category is

the most frequently identified class. This pattern indicates substantial overlap in the financial profiles of firms near the middle of the tax-risk distribution. Additional firm-level information, such as more detailed tax-footnote disclosures or related-party transaction characteristics, may therefore be required to improve classification performance for non-extreme cases.

The contrasting effects of hyperparameter optimisation are also noteworthy. Optimisation improved the Random Forest across all major metrics but reduced the performance of the SVM. This asymmetry demonstrates that hyperparameter optimisation is model-dependent and should be evaluated empirically rather than assumed to improve every algorithm.

6.3 Firm Heterogeneity and Clustering

The clustering results reveal that the majority of Korean listed non-financial firms share broadly similar financial and governance characteristics, while a smaller subset exhibits more distinctive profiles. The GMM profiles identify firm size, foreign ownership, cash holdings, and asset tangibility as important dimensions of differentiation.

The probabilistic structure of GMM provides a useful complementary view of the data because it can accommodate overlapping firm profiles rather than forcing every observation into a strictly separated spherical cluster. The clustering results should therefore be interpreted as descriptive segmentation rather than official economic classifications.

6.4 Explainability and Financial Interpretation

The SHAP analysis provides an additional layer of interpretation for the final LightGBM regression model. The dominance of A_GETR, GETR5, CETR5, and related tax-history variables indicates that previous tax behaviour is highly informative for future CETR prediction. Traditional financial variables such as ROE, GRW, and PPE provide additional predictive information but are secondary to the historical tax variables.

These findings reinforce the importance of temporal information in corporate tax analysis. Importantly, SHAP values describe the contribution of variables to the model’s predictions and should not be interpreted as evidence that a variable causally determines corporate tax avoidance.

6.5 Methodological Implications

From a methodological perspective, the study demonstrates the importance of temporal evaluation when analysing corporate panel data. Chronological training and testing periods, combined with fit-on-train-only preprocessing, provide a more realistic representation of a forecasting setting than random observation-level splitting.

The results also indicate that feature engineering may be more important than simply increasing model complexity. The optimised LightGBM substantially improved when informative historical tax variables were added, whereas the Residual MLP remained inferior to LightGBM despite

its nonlinear architecture. This suggests that the quality, timing, and relevance of the input variables are central to predictive performance.

6.6 Limitations

Several limitations should be noted. First, the predictive performance of the models remains moderate, reflecting the inherent difficulty of forecasting a proxy measure that is itself an indirect representation of latent tax avoidance behaviour [8]. Second, the study uses CETR as the primary regression target; extending the framework to other tax-avoidance proxies could provide a broader assessment of tax behaviour. Third, the leakage-free framework excludes the final observation year for each firm because no subsequent target is available, which may affect the composition of the predictive sample. Fourth, the clustering analysis is descriptive and should not be interpreted as causal or definitive firm classification.

7 Practical Applications and Implications

The findings of this study have several practical implications for tax authorities, corporate managers, auditors, investors, and financial analysts.

For tax authorities, the predictive framework provides a potential analytical approach for screening firms according to expected future tax-risk levels. The one-year-ahead design allows currently available financial and tax information to be used to identify firms that may warrant additional analytical attention. The resulting scores should be treated as screening information rather than a substitute for formal tax audits or regulatory investigations.

For corporate managers and tax professionals, the strong contribution of historical tax-related variables suggests that the firm’s own tax history can be incorporated into internal monitoring and risk-assessment processes. Tracking the persistence of effective tax-rate measures may help organisations identify periods in which tax behaviour changes materially from historical patterns.

For auditors and investors, the classification and regression outputs provide complementary information. Regression produces a continuous forecast of future CETR, while classification produces discrete risk categories that may be easier to incorporate into screening procedures. The combination can support broader financial and governance analysis.

The clustering results also provide a basis for firm segmentation. Firms can be examined according to combinations of size, foreign ownership, leverage, cash holdings, and asset structure rather than relying exclusively on a single financial measure. Finally, SHAP explainability provides a more transparent interpretation of model outputs by identifying the variables that contribute most strongly to individual predictions and to the overall model behaviour.

8 Conclusion

This study examined corporate tax avoidance among Korean listed companies using a comprehensive machine learning framework based on the KoTaP dataset. The results show that nonlinear machine learning models, particularly the optimised LightGBM model, provide better one-year-ahead CETR prediction than the linear baseline, while the optimised Random Forest achieves the strongest overall classification performance.

The SHAP analysis further shows that historical tax-related variables are among the most influential contributors to future CETR predictions, supporting the presence of strong temporal persistence in corporate tax behaviour. The clustering analysis additionally reveals meaningful differences in the financial characteristics of firms, particularly in relation to size, foreign ownership, cash holdings, leverage, and asset structure.

Overall, the study demonstrates that combining prediction, classification, clustering, hyperparameter optimisation, and explainability provides a broader understanding of corporate tax avoidance than relying on a single analytical technique. Nevertheless, the moderate predictive performance indicates that important determinants remain outside the available financial variables. Future research can extend the framework using additional tax-avoidance measures, longer forecasting horizons, richer corporate disclosures, and other sources of financial and institutional information.

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